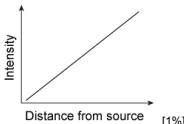
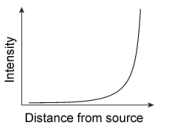
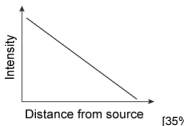
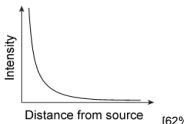
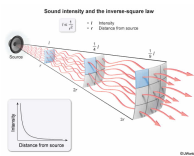


Which of the following graphs best illustrates how the intensity of sound varies as a detector is moved further and further from a stationary source projecting sound in all directions? (Note: Ignore the Doppler effect)

- ☐ A.
- 
- [1%]
- ☐ B.
- 
- ☐ C.
- 
- [35%]
- ☒ D.
- 
- [62%]

Correct
63% answered correctly

Explanation:



Sound waves propagate energy by increasing and decreasing the pressure within **consecutive segments** of a medium through which they travel. The pressure gradient caused by sound wave propagation is related to the **intensity (*I*)** of the sound wave, which is defined as **power (*P*)** per unit **area (*A*)**:

$$I = \frac{P}{A}$$

Although the power of sound waves is largely conserved when passing through a uniform medium (eg. air), the intensity of sound waves (and perceived loudness) is not. This phenomenon can be explained by the increased area onto which sound waves impart energy as the distance between source and detector increases. In the case of a single source of sound propagating sound waves in all directions, increasing area is best illustrated by a series of spheres with increasing radius (*r*) and **surface area (*A*)** as given by:

$$A = 4 \pi r^2$$

Substituting this equation into the equation relating intensity, power, and area yields:

$$I = \frac{P}{4 \pi r^2} = \frac{P}{4 \pi} \cdot \frac{1}{r^2}$$

Simplifying the above equation as a proportionality to isolate the relationship between intensity and radius (distance) results in:

$$I \propto \frac{1}{r^2}$$

Therefore, the intensity of sound projected in all directions is inversely proportional to the square of the distance between the source and the detector.

(Choices A and B) For intensity to increase as distance between sound source and observer increases, the area onto which sound imparts energy would have to decrease. The reflection and scattering of sound makes such a scenario unlikely in everyday circumstances.

(Choice C) Although the relationship between distance and sound intensity is inverse, the relationship is exponential, **not linear**.

Educational objective:

The intensity of sound waves is defined as power per unit area, and relates to the perceived loudness of sounds. Intensity is inversely proportional to the square of the distance between source and detector, resulting in a relatively significant decrease in intensity as the distance increases.

Time spent:0 sec
Subject:
Physics

QID:402128
System:
Light and Sound

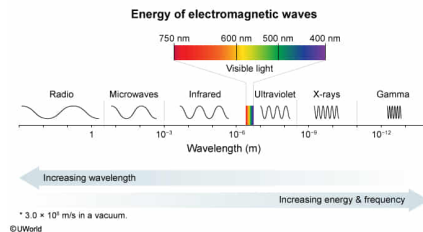
Which of the following color transitions best describes what may take place as a light-emitting object begins releasing electromagnetic waves of progressively higher energy?

- ☐ A. Blue to green to red [16%]
- ✓ ☒ B. Red to green to blue [79%]
- ☐ C. Blue to red to green
- ☐ D. Red to blue to green [2%]

Correct

79% answered correctly

Explanation:



Electromagnetic radiation refers to the propagation of waves of oscillating electric and magnetic fields. The **electromagnetic spectrum** refers to all classifications of radiation, which differ from one another in terms of **frequency** (f), **wavelength** (λ), and energy but are related to one another in terms of the **speed of light in a vacuum** ($c = 3 \times 10^8$ m/s):

$$c = \lambda f$$

The complete electromagnetic spectrum is often visually represented by an array of radiation organized from radio waves (largest wavelength and lowest frequency) to gamma rays (smallest wavelength and highest frequency). The subspectrum of **visible light** is featured within the middle of this array, spanning the range of wavelengths from about 750 nm (red) to approximately 400 nm (violet).

Not only does each color of visible light have a different wavelength and frequency, but each also has a different energy. This energy is described by the Planck-Einstein equation, which relates the **energy** (E) of an electromagnetic wave (and corresponding photon) to wave frequency and Planck's constant ($h = 6.62 \times 10^{-34}$ m²·kg/s):

$$E = hf$$

Combining both of the previous equations gives an expression for the energy with respect to wavelength:

$$E = \frac{hc}{\lambda}$$

In the question, a light-emitting object releases electromagnetic waves of progressively higher energy. Based on the equations above, the frequency of these emitted electromagnetic waves must progressively increase and the wavelength must progressively decrease. Accordingly, when emitting across the visible light spectrum from lower to

higher energy, the **color progression** would be red, orange, yellow, green, blue, indigo, and violet. Therefore, the transition is from **red light** to **green light** to **blue light**.

Educational objective:

The visible light spectrum refers to electromagnetic radiation that may be detected by the human eye, ranging from approximately 750 to 400 nm in wavelength (ie, red to violet). The energy associated with electromagnetic radiation is directly proportional to wave frequency and inversely proportional to wavelength.

Time spent: 0 sec

Subject:
Physics

QID: 402129

System:
Light and Sound

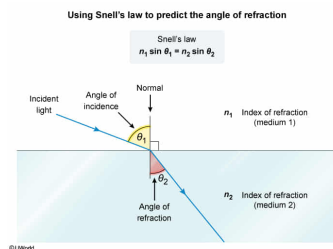
Relative to the angle of incidence, how will angle of refraction (θ_2) change when a monochromatic ray of light enters a medium of greater refractive index?

- ☐ A. θ will increase, because total internal reflection occurs. [10%]
☐ B. θ will not change, because only multi-color light refracts. [2%]
☐ C. θ will decrease, because the energy of light decreases in the new medium. [13%]
☒ D. θ will decrease, because light travels slower in the new medium. [74%]

Correct

74% answered correctly

Explanation:



Refraction refers to the bending of light that takes place as light traveling at an angle passes from one medium to the next. Light will either bend (refract) toward or away from the **normal** (an axis perpendicular to the interface between media) depending on the **index of refraction (n)** of each medium.

Snell's law relates the index of refraction of each medium (n_1 and n_2) to the **angle of incidence (θ_1)** and the **angle of refraction (θ_2)**, the angle at which light approaches and departs the interface between media, respectively:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Rearrangement of Snell's law yields a form of the expression that is useful in predicting the direction of refraction when the relationship between refractive indices is known:

$$\frac{n_1}{n_2} = \frac{\sin \theta_2}{\sin \theta_1}$$

In this question, light leaves a medium with a lower refractive index and enters a medium with a higher refractive index such that:

$$\frac{n_1}{n_2} = \left[\frac{\text{Lower}}{\text{Higher}} \right] \leq 1$$

Therefore, $\sin \theta_2 / \sin \theta_1$ must also be less than 1:

$$\frac{\sin \theta_2}{\sin \theta_1} \leq 1$$

Consequently, θ_1 must be **greater than** θ_2 as light enters a medium of higher refractive index. In other words, the angle of refraction will decrease relative to the angle of incidence (ie, light will bend toward the normal) (**Choice D**).

(Choice A) **Total internal reflection** may only occur when light leaves a medium with a higher refractive index and enters a medium with a *lower* refractive index, which is the opposite case to this scenario.

(Choice B) Optical phenomena are often modeled by experiments utilizing monochromatic light (containing only one color). However, *all* forms of light are subject to refraction.

(Choice C) The energy of **light** is a product of light frequency and Planck's constant, *not* the medium through which light travels.

Educational objective:

Snell's law may be used to predict the relationship between the angle of incidence and angle of refraction provided that the relationship between the refractive index of each medium is known. Light will bend toward the normal when it passes through a media interface from lower to higher refractive index.

Time spent: 0 sec
 Subject:
 Physics

QID: 402130
 System:
 Light and Sound